

NMR Features of Abinit: Relativistic Terms, Spin Couplings, and Spatial Maps

J. W. Zwanziger

Department of Chemistry, Dalhousie University, Halifax, NS B3H 4R2 Canada

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Magnetic shielding in NMR

Total energy:

$$E = \underbrace{-\mu \cdot \mathbf{B}}_{\text{Zeeman}} + \underbrace{\mu \cdot \overleftrightarrow{\sigma} \cdot \mathbf{B}}_{\text{shielding}}$$

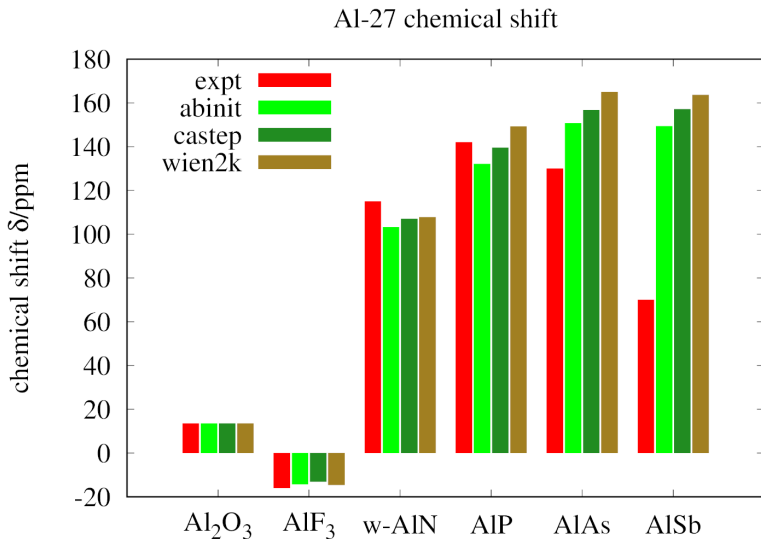
$$\mu \cdot \underbrace{\overleftrightarrow{\sigma} \cdot \mathbf{B}}_{\text{induced field}}$$

$$\underbrace{\mu \cdot \overleftrightarrow{\sigma}}_{\text{induced dipole}} \cdot \mathbf{B}$$

- Standard approach
- As done by GIPAW: estimate current due to external $\mathbf{B}$, use Biot-Savart law to compute induced $\mathbf{B}$
- “Converse” method
- Apply nuclear magnetic dipole by hand
- Compute orbital magnetization to determine induced dipole. Fully implemented in ABINIT, see `orbmag` keyword

Zwanziger, Torrent, Gonze, PRB 107, 165157 (2023)

Results were good, but lacking for heavy nuclei



The Dirac equation is approximated to zeroth order, leading to a correction factor $\mathcal{K} = (1 + \frac{1}{2}\alpha^2 V)^{-1}$, and effective Hamiltonian

$$H = \frac{1}{2}\vec{\sigma} \cdot (\vec{p} + \vec{A}_0 + \vec{A}_\mu)\mathcal{K}\vec{\sigma} \cdot (\vec{p} + \vec{A}_0 + \vec{A}_\mu) + V,$$

where $\vec{\sigma}$ are the Pauli spin matrices (electron spin).

Because of the derivatives ($\vec{p} = -i\nabla$) and the spin degrees of freedom, many terms arise.

Relevant resulting terms

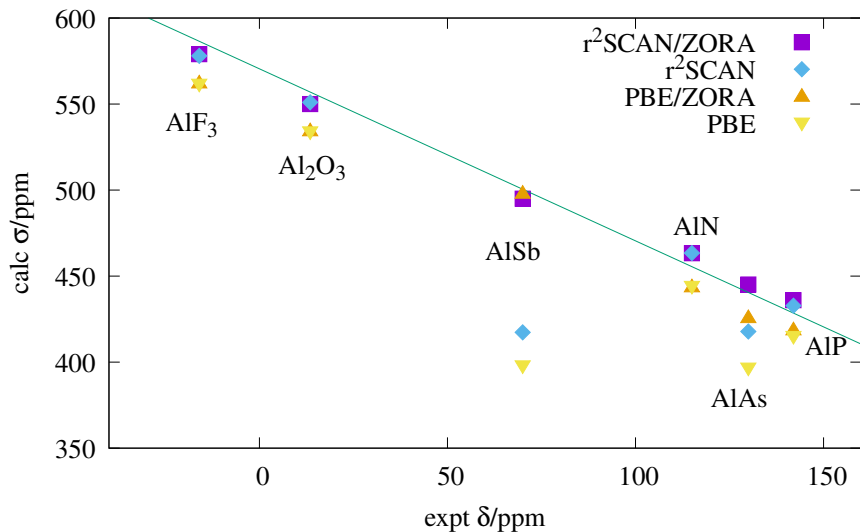
Effective electronic Hamiltonian (see new zora keyword): ¹

$$H = \underbrace{\frac{1}{2}(\vec{p} + \vec{A}_0 + \vec{A}_\mu)^2}_{\mathcal{K} \approx 1 \text{ away from nuclei}} + V + \underbrace{\mathcal{K}' |\vec{r}_R|^{-1} \vec{L}_R \cdot \vec{S}}_{\text{spin-orbit}} - \underbrace{\mathcal{K} \alpha^2 |\vec{r}_R|^{-3} \vec{m} \cdot (\mathbb{1} - 3\hat{r}\hat{r}) \cdot \vec{S} + \alpha^2 \mathcal{K}' |\vec{r}_R|^{-2} \vec{m} \cdot (\mathbb{1} - \hat{r}\hat{r}) \cdot \vec{S}}_{\text{electron spin-nuclear dipole}}.$$

(note $\vec{r}_R = \vec{r} - \vec{R}$)

¹Zwanziger, Farrant, Werner-Zwanziger, *J. Magn. Reson.* **374**, 107861 (2025)

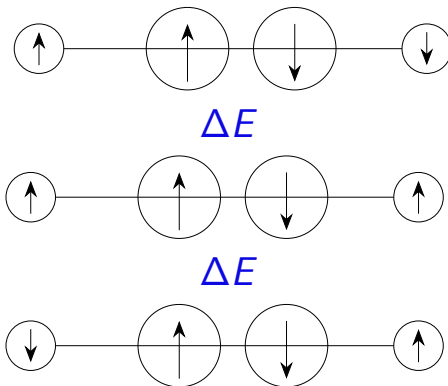
Results for Al-27 Shifts



Slope -1; intercept 570 ppm [600 ppm, Antušek and Holka *JCP* **143**, 074301, (2015)]

Through-bond coupling (schematic)

Assume $H = J\vec{\mu} \cdot \vec{S}$ with $J > 0$



Through-bond coupling (formal)

Defining Hamiltonian term:

$$\mathbf{J}_{\alpha,\beta}^{ij} = \frac{\partial^2 H}{\partial \vec{\mu}_{\alpha}^i \partial \vec{\mu}_{\beta}^j}$$

Perturbative treatment, used in most codes:

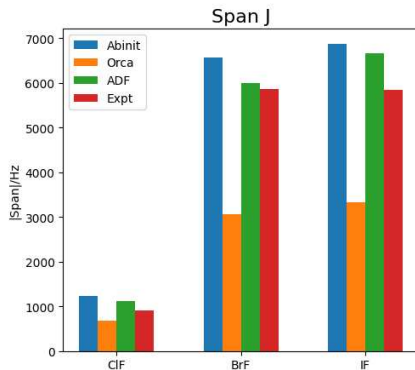
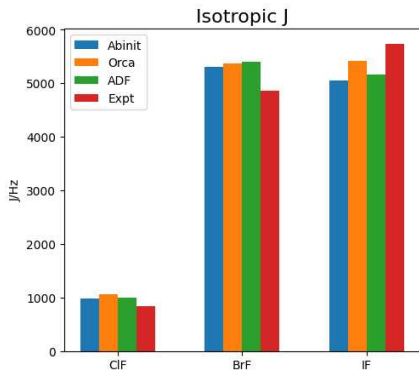
$$\mathbf{J}_{\alpha,\beta}^{ij} = \langle \psi^{(0,0)} | H^{(1,1)} | \psi^{(0,0)} \rangle + 2 \operatorname{Re} \langle \psi^{(0,0)} | H^{(1,0)} | \psi^{(0,1)} \rangle,$$

If the effect of the nuclear dipole is explicitly computed, then $\mathbf{J}_{\alpha,\beta}^{ij}$ can be computed *non-perturbatively*, from a finite-difference scheme of total energies

$$\mathbf{J}_{\alpha,\beta}^{ij} = \frac{1}{4} \left[E(+\vec{\mu}_{\alpha}^i, +\vec{\mu}_{\beta}^j) - E(-\vec{\mu}_{\alpha}^i, +\vec{\mu}_{\beta}^j) \right. \\ \left. - E(+\vec{\mu}_{\alpha}^i, -\vec{\mu}_{\beta}^j) + E(-\vec{\mu}_{\alpha}^i, -\vec{\mu}_{\beta}^j) \right]$$

Results on molecules

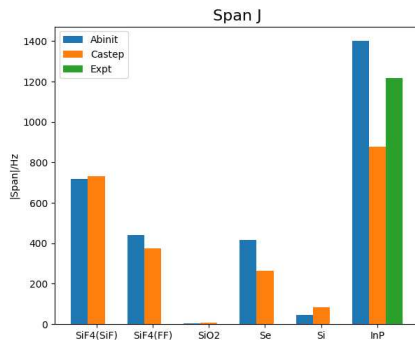
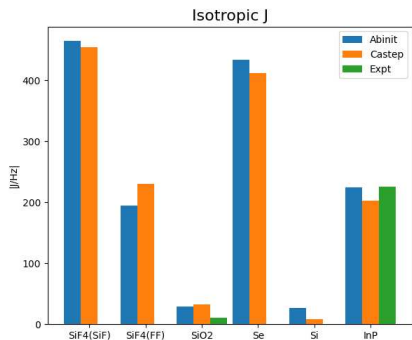
Code comparisons with experiment.



Zwanziger, SSNMR 141, 102060 (2026); Bryce and Wasylshen, JACS 122, 3197 (2000)

Results on solids

Code comparisons with experiment.



Zwanziger, SSNMR 141, 102060 (2026); Tomaselli *et al.*, PRB 58, 8627 (1998); Georgiou, Cuniberti, Bouchard *et al.*, Nature Commun. 15, 7367 (2024); Perras, Zwanziger, Rossini, Nature Commun. 16, 10032 (2025)

Interpreting the shielding: gauge invariance

Detailed interpretation of the shielding is fraught with problems. Consider:

$$\begin{aligned}\sigma &= \sigma^d + \sigma^p \\ \sigma_{xy}^d &= \sum_v^{\text{occ}} \langle \psi_v | \frac{r_O r_N \delta_{xy} - r_{Ox} r_{Ny}}{r_N^3} | \psi_v \rangle \\ \sigma_{xy}^p &= \sum_c^{\text{empty}} \frac{\psi_c | L_O | \psi_{\text{occ}} \rangle \langle \psi_{\text{occ}} | L_N / r_N^3 | \psi_c \rangle}{E_0 - E_c}\end{aligned}$$

- σ^d and σ^p depend on gauge origin r_O for magnetic field, though total σ does not.
- Total σ is invariant to unitary transformation of occupied $|\psi\rangle$ orbitals.

Consequences and what to do

- Separation into σ^d and σ^p is not well posed and not comparable among different gauge choices.
- Projection of σ onto different orbital and band contributions is not well posed, and also not comparable amongst different orbital choices.
- Consider interpretation of total shielding density, expressed in real space.²

²C. Jameson and A. D. Buckingham, *J Chem Phys* **73**, 5684 (1980).

Shielding and current density

The shielding is computed by a sum over (many!) matrix elements

$$\langle u' | H | u \rangle$$

. Planewave basis expansion:

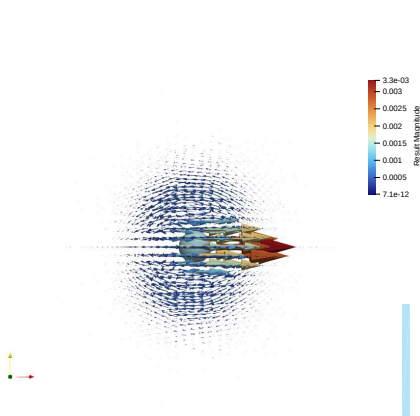
$$\sum_{G', G} \underbrace{\langle u' | G' \rangle \langle G' | H | G \rangle \langle G | u \rangle}_{T_{G', G}},$$

where $\langle \mathbf{r} | \mathbf{G} \rangle = \exp(i\mathbf{r} \cdot \mathbf{G})$. Now Fourier-transform T over G' and G to obtain shielding as a density in real space.

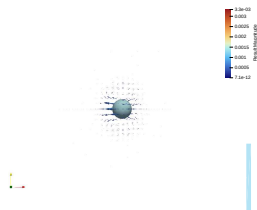
Currents can also be obtained, by relating the induced dipole $\vec{m}\sigma$ to the current density through

$$\vec{m}\sigma = \frac{1}{2} \vec{r} \times \vec{j}(\vec{r}).$$

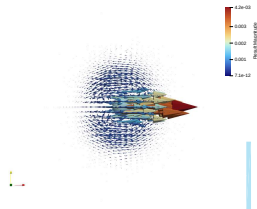
Neon atom shielding density



All terms

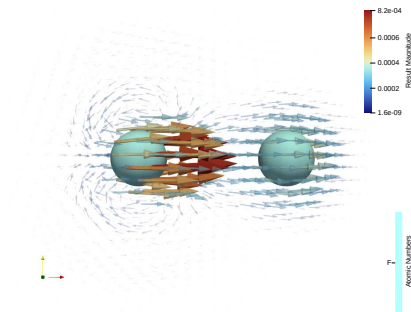


Para-like

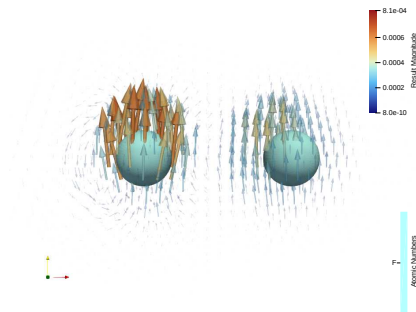


Dia-like

F_2 shielding density

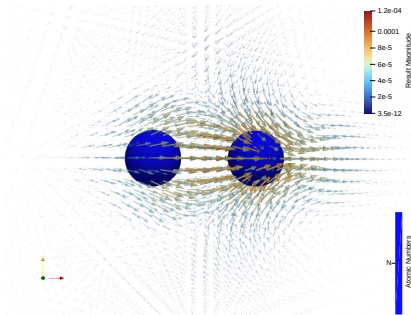


Dipole parallel to bond

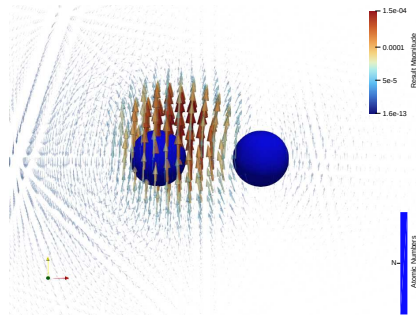


Dipole perpendicular to bond

N_2 shielding density

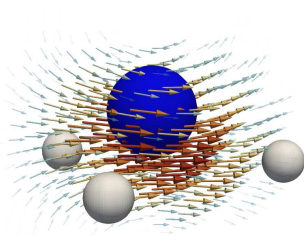


Dipole parallel to bond

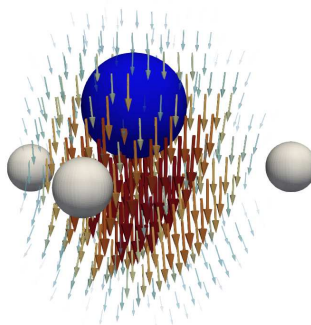


Dipole perpendicular to bond

NH_3 shielding density

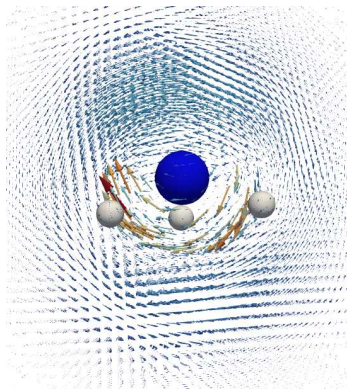


Dipole in x direction

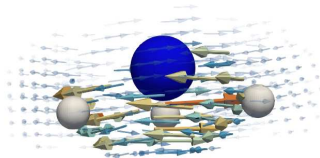


Dipole in z direction

NH_3 induced current density



Dipole in x direction



Dipole in z direction

Multiple NMR-related updates, including

- New ZORA-derived terms, required for accuracy for heavier nuclei
- Implemented and tested scheme for non-perturbative J-coupling
- Implementing real-space density maps for shielding and induced current

Thanks as always to the collective, especially Xavier, Marc T, Matthieu, and Matteo!